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SALOME: Bridging CAD, Meshing, and Visualization in Numerical Simulations

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Summary

SALOME [1] is an open-source platform that offers researchers and engineers a comprehensive environment for numerical simulation pre- and post-processing tasks. Primarily developed by the French Alternative Energies and Atomic Energy Commission (CEA) and Électricité de France (EDF), SALOME is composed of different modules that provide: Computer Aided Design (CAD), meshing, mesh adaptation, data visualization, calculation supervision, and uncertainty quantification services. It supports complete computational workflows across a wide range of scientific and engineering domains, including computational fluid dynamics (CFD), structural mechanics, thermal analysis, and multiphysics coupling. Users can interact with these modules either through an intuitive graphical user interface (GUI) or through Python scripts, with full interoperability between both modes of operation. This dual-interface approach improves accessibility for novice users while providing flexibility for advanced practitioners, and enables efficient automation of complex workflows. A key feature of SALOME is its extensibility, allowing researchers and software developers to construct domain-specific applications by assembling and customizing its modules. This capability facilitates the development of tailored solutions for particular numerical simulation challenges, such as SALOME_CFD (Yvan et al., 2021) for fluid dynamics, SALOME_MECA (code_aster, 2024) for structural mechanics, and other domain-specific platforms that integrate custom solvers, visualization pipelines, or coupling strategies.

Statement of need

Numerical simulations are widely used in science and engineering to study complex physical phenomena and optimize system designs. A typical simulation workflow involves several key steps: computational geometry creation or import, mesh generation, physical model setup, numerical solution, and post-processing the results for analysis. Each of these steps requires specialized tools and expertise. While powerful specialized tools exist for individual steps, integrating them into a seamless, automated pipeline remains challenging. Data exchange issues, inconsistent user interfaces, and limited scripting capabilities often lead to time-consuming manual work and reduced reproducibility. SALOME was developed to address this fragmentation by offering a unified, extensible open-source platform, hosting all these services under one roof.

[1] SALOME stands for *Simulation numérique par Architecture Logicielle en Open source et à Méthodologie d'Évolution*, which translates to *Numerical Simulation using Open Source Software Architecture with Evolutionary Methodology*.

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State of the Field

For numerical simulations, commercial platforms such as ANSYS, Abaqus, COMSOL Multiphysics, offer integrated environments but are constrained by high licensing costs and limited customizability. In the open-source community, powerful individual tools have emerged, such as FreeCAD for parametric CAD, Gmsh and Netgen for meshing, ParaView for visualization, and solvers such as OpenFOAM, MOOSE, and FreeFEM. These components are powerful and widely adopted, yet combining them into automated workflows remains cumbersome due to interoperability and interface issues. SALOME addresses this gap and provides a tightly integrated open-source alternative. The individual modules of SALOME are described in detail in the following section.

Software Design

SALOME is built on a modular architecture offering a comprehensive suite of tools for the entire simulation workflow.

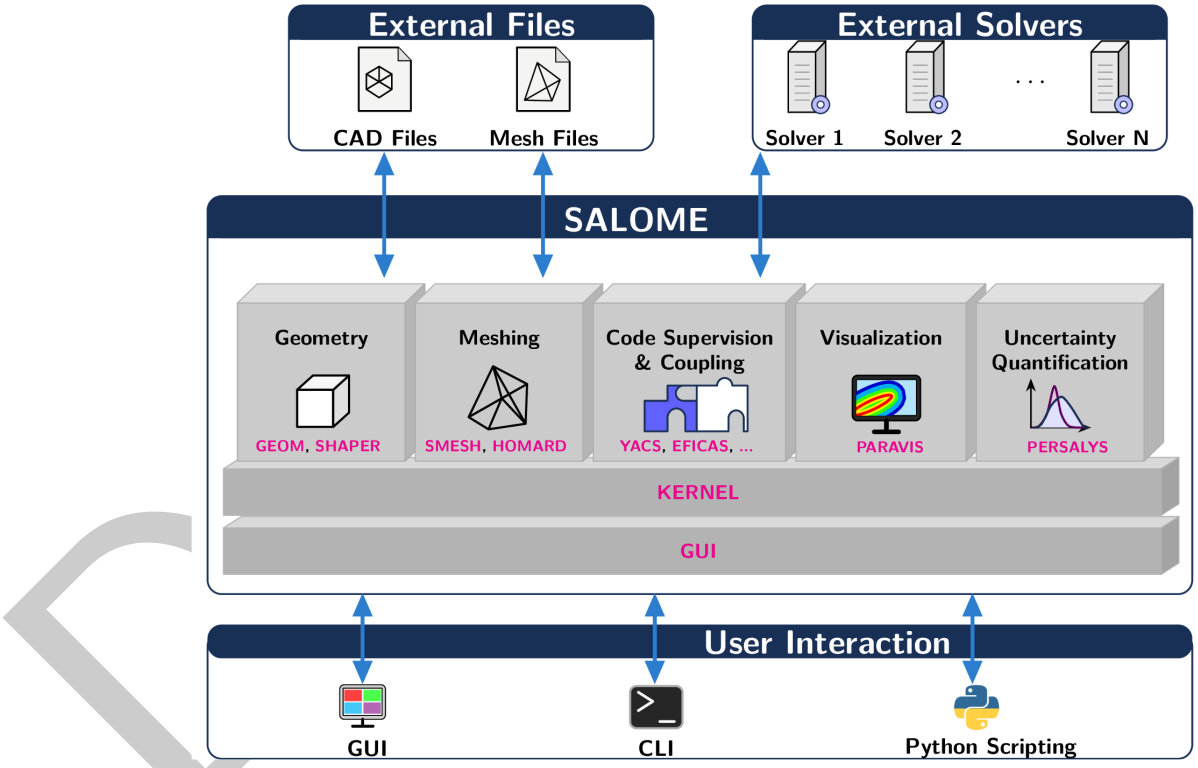


Figure 1: Modular architecture of the SALOME platform, with SALOME modules highlighted in magenta.

This modularity is a deliberate design choice in SALOME over a monolithic, single-purpose tool. Each module (CAD, meshing, visualization, etc.) is developed and maintained largely independently, each in its own repository. This makes it easier to add, update, remove, or replace individual capabilities of a module over time. Two foundational modules, KERNEL and GUI, provide the common infrastructure that brings these independently developed modules together into a single consistent platform. KERNEL manages the underlying application services and module integration, while GUI provides a unified Qt-based graphical interface through which users can access and interact with the available modules. At the scripting level, a common Python API lets objects (geometries, meshes, fields) created in one module be

passed directly to another, so a workflow can move from CAD to meshing to visualization without manual data conversion between modules. Each module is found at runtime through environment variables lookup that dynamically builds the application with the selected modules. Figure 1 illustrates this architecture, highlighting how the independently developed functional modules are brought together through the common KERNEL and GUI infrastructure and interact with external files, solvers, and user interfaces. The individual modules that make up this architecture are described in detail below.

CAD

SHAPER and **GEOM** modules provide CAD capabilities built on top of (*Open CASCADE Technology*, 2024). While GEOM focuses on procedural, scripting-oriented geometry creation, SHAPER offers a modern, parametric, feature-based GUI. Key functionalities include:

- Interactive or Python-scripted creation of complex geometries.

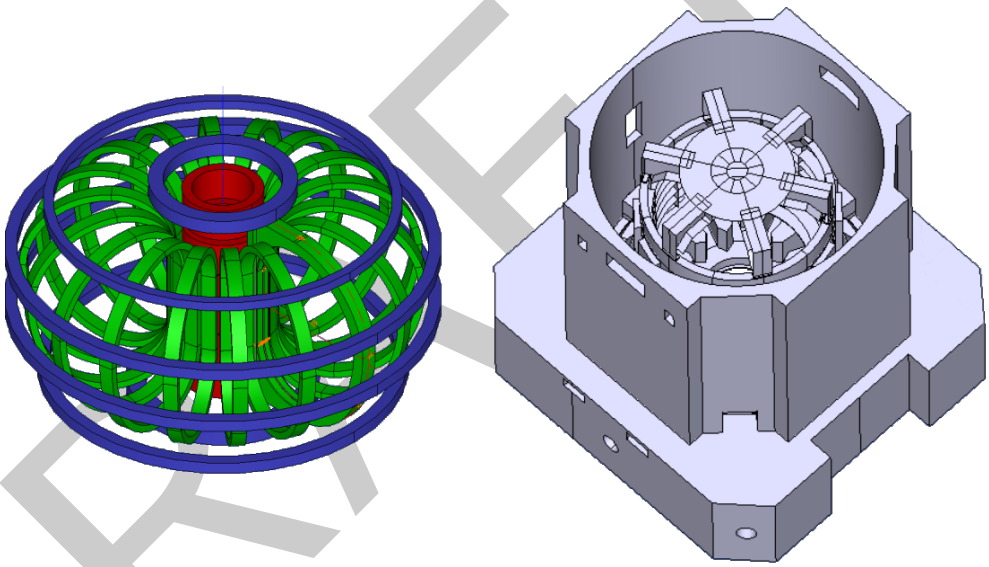


Figure 2: Left: magnet system of a nuclear fusion synchrotron (tokamak) geometry detailed in GEOM, from (Quentin Le Coz, 2019), right: complex geometry detailed with SHAPER.

- Import/export for standard CAD formats (e.g., XAO, STEP, IGES, STL) to ensure interoperability.
- Advanced geometry cleaning, healing, and Boolean operations.
- Parametric modeling for easy geometry updates and parametric studies.

Figure 2 displays industrial geometries generated by these modules, including a tokamak component constructed in GEOM and complex CAD in SHAPER.

Mesh

SMESH is the meshing module of SALOME designed to produce meshes suitable for numerical simulation. For mesh generation, SMESH incorporates SALOME's in-house meshing algorithms, alongside open-source ones from Gmsh (Geuzaine & Remacle, 2009) and NETGEN (Schöberl, 1997), and commercial ones (MeshGems) from (Spatial, 2024). SMESH provides:

- High-quality 0D–3D mesh generation, including high-order elements.
- Diverse element types (hexahedral, tetrahedral, polyhedral) and hybrid meshes.
- Algorithm mixing for versatile handling of complex geometry.

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- Quality analysis, automatic correction, and extensive format support (e.g., MED, CGNS).

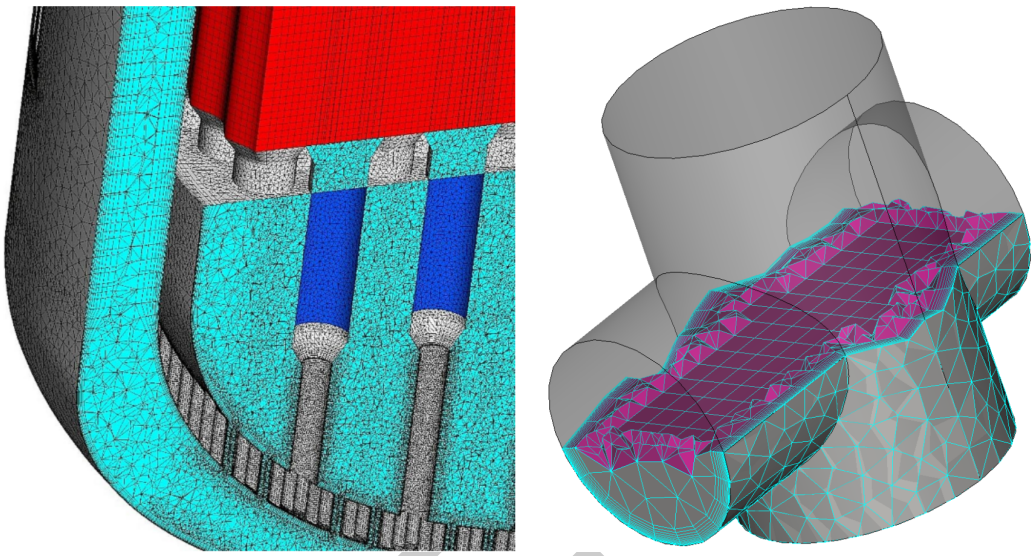


Figure 3: Left: hybrid mesh of a nuclear reactor component, from (Böttcher et al., 2024), right: hexahedral-core mesh.

Figure 3 illustrates SMESH's interoperability, combining algorithms like MeshGems, native viscous layers, and extrusion to mesh complex reactor components.

HOMARD performs mesh adaptation based on solution fields and refinement strategies, supporting iterative workflows (Figure 4).

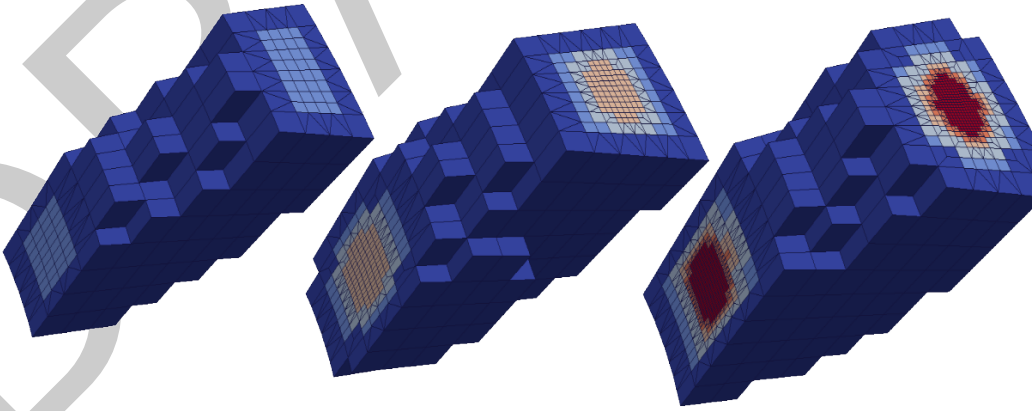


Figure 4: An example of mesh adaptation in HOMARD.

MEDCOUPLING handles mesh/field data exchange, parallel interpolation, and co-simulation. Built on the MED format (a standardized data model for mesh and field data), it ensures interoperability between multiphysics codes (Figure 5).

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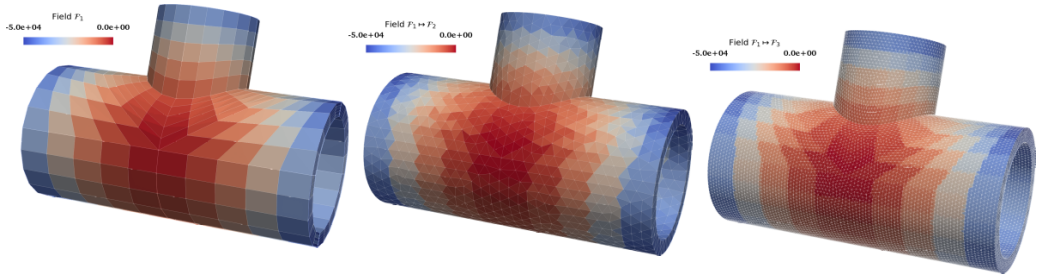


Figure 5: An example of mesh field interpolation using MEDCOUPLING.

Simulation Workflow and Supervision

YACS is SALOME's module for coordinating multidisciplinary simulations involving the interaction of multiple physical models or codes. It allows users to design, automate, and control simulation workflows through a graphical interface or Python scripts, and supports both sequential and coupled execution strategies.

EFICAS is a module in SALOME that serves to generate multi-code inputs and environments. It handles multiple versions of supported codes through developer-defined "Catalogue" files that specify available commands and parameters.

JOBMANAGER is a module for managing the execution of simulation jobs on local or remote computing resources. It provides functionalities for job submission, monitoring, and resource allocation, streamlining the integration of simulation workflows with batch or HPC environments.

Visualization

PARAVIS is SALOME's integrated module for advanced data visualization, built on ParaView (Ayachit, 2015). It is accessible through both the graphical interface and Python scripts, and is extended with specialized plugins for numerical simulation post-processing. Notable features include visualization of mechanical fields at quadrature points, support for loading MED files, a static-mesh filter to accelerate post-processing of transient simulations, and tools for remeshing slices into triangular meshes. Figure 6 presents two examples of simulation results visualized with PARAVIS.

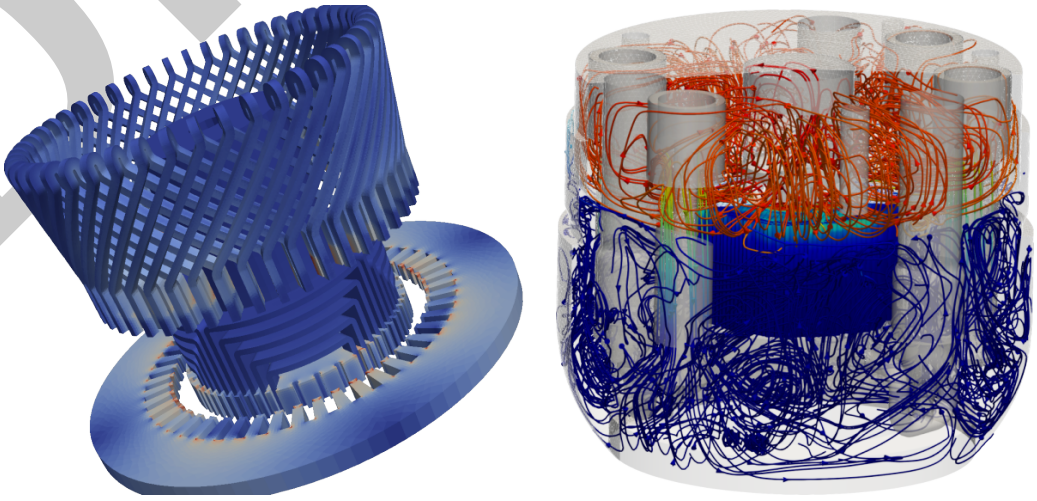


Figure 6: Two examples of numerical solution field visualizations in PARAVIS.

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Uncertainty Quantification

PERSALYS, the graphical user interface of OpenTurns (Airbus et al., 2025), is a native SALOME module developed by EDF for uncertainty quantification (UQ), sensitivity analysis, surrogate modeling, and optimization. It provides an intuitive interface for defining input uncertainties, building surrogate models, performing global sensitivity analyses, and managing optimization workflows. It enables robust parametric studies and uncertainty propagation through simulation chains with minimal setup effort.

Research Impact Statement

SALOME is widely used in academia and industry, supporting research and development across multiple engineering disciplines. Below is a non-exhaustive list of recent solvers and platforms that integrate SALOME:

- code_aster: state-of-the-art finite element solver for mechanics, relying on SALOME for geometry, meshing, and post-processing via SALOME_MECA (code_aster, 2024).
- code_saturne: (code_saturne, 2025) a parallel finite-volume CFD solver integrated into SALOME_CFD (Yvan et al., 2021).
- Kratos Multiphysics: a parallel, multi-disciplinary FEM framework (Vicente Mataix et al., 2024) ported to SALOME (Philipp et al., 2021).
- AZTLAN platform: a Mexican platform for nuclear reactor analysis and design (Torres et al., 2015).
- DRAGON5/DONJON5: (Hébert, 2013) platforms for fission reactor simulation, including space applications, integrated into SALOME(Hébert, 2014).
- ALAMOS: a geometry and meshing tool for neutron physics in nuclear reactors (Tomatis et al., 2022).
- SMARDDA: a plasma–surface interaction module for tokamak design using SALOME for GUI, CAD, meshing, and visualization (Kos et al., 2019).
- McCAD: a geometry conversion tool for Monte Carlo simulations that uses SALOME for CAD manipulation and material assignment (Harb et al., 2023).

SALOME has also been used as a component in numerical simulations reported in recent publications covering FEM–FVM coupling (Barbi et al., 2021; Zerkak et al., 2007), molecular dynamics (Biagooi & Nedaaee Oskoee, 2020), nuclear engineering (Aydemir et al., 2019; Zhang et al., 2021), aerospace engineering (Ermakov & Kryuchkova, 2020), and magnet design (Nunio & Manil, 2011; Pirapakaran et al., 2023).

Documentation and Availability

Comprehensive tutorials and detailed user and developer documentation are available for all SALOME modules (*Salome Platform Documentation*, 2024).

SALOME provides precompiled binaries for Linux and Windows, along with the SAT tool for building from source. Ready-to-use binaries are available from the official SALOME website (https://www.salome-platform.org/).

AI Usage Disclosure

AI tools were used for minor editorial support and language polishing during the preparation of this manuscript. All software-related aspects, including source code, development, design, testing, and validation, were carried out and verified by human developers and contributors.

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